PROPER PUSHFORWARD COMMUTES WITH THE K-CYCLE CLASS OF A COHERENT SHEAF WHOSE SUPPORT HAS DIMENSION AT MOST K

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's parenthetical at source line 2040, 'The case of a closed immersion is straightforward (omitted).', is retained verbatim inside the counted proof; the closed-immersion case is the routine residue of an argument whose main reduction and final multiplicity comparison are fully carried out in the span.

Allowed standard prerequisites (editorial).

- Cycle class of a closed subscheme versus that of its structure sheaf. Same chapter chow.tex, label lemma-cycle-closed-coherent, invoked at proof line 2015. Reduces part (1) of the statement to part (2) applied to the structure sheaf O_Z; part (1) is therefore not counted separately and is settled by the cited lemma plus the argument in the span.
- Every coherent sheaf is the pushforward of a coherent sheaf on the closed subscheme given by its support. Cohomology of Schemes, labels coherent-lemma-coherent-support-closed (proof line 2018) and coherent-proposition-proper-pushforward-coherent (used in the statement at proof lines 2008-2011). Produces the closed immersion i : Z -> X and the coherent O_Z-module G with i_* G isomorphic to F, which is the first reduction of the proof.
- Finiteness of the generic fibre and the pushdown formula for a finite ring map. Morphisms, labels morphisms-lemma-quasi-compact-scheme-theoretic-image (proof line 2042) and morphisms-lemma-generically-finite (proof line 2069); same chapter chow.tex, label lemma-multiplicity-finite (proof line 2066); Algebra, label algebra-lemma-pushdown-module (proof line 2094). Makes f|_Z dominant and, after shrinking to an open neighbourhood of the generic point, finite and affine, and finally converts the coefficient of Z

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in f_*[F]_k into the sum over the primes of A over p via the pushdown formula.

Source reference. Original lemma 02M0: sources/stacks-chow/tags.txt, source lines 3427–3427. Navigation pointer for \ref{algebra-lemma-pushdown-module} used inside the counted span (line 2094); the argument that consumes it is retained in the span.

Source reference. Original lemma 01Y5: sources/stacks-chow/tags.txt, source lines 2544–2544. Navigation pointer for \ref{coherent-lemma-coherent-support-closed} used inside the counted span (line 2018); the argument that consumes it is retained in the span.

Source reference. Original lemma 02O5: sources/stacks-chow/tags.txt, source lines 3504–3504. Navigation pointer for \ref{coherent-proposition-proper-pushforward-coherent} used inside the counted span (line 2011); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QY: sources/stacks-chow/pinned-original-source.tex, source lines 1776–1776. Navigation pointer for \ref{lemma-cycle-closed-coherent} used inside the counted span (line 2015); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QT: sources/stacks-chow/pinned-original-source.tex, source lines 1629–1629. Navigation pointer for \ref{lemma-multiplicity-finite} used inside the counted span (line 2066); the argument that consumes it is retained in the span.

Source reference. Original lemma 02NW: sources/stacks-chow/tags.txt, source lines 3495–3495. Navigation pointer for \ref{morphisms-lemma-generically-finite} used inside the counted span (line 2069); the argument that consumes it is retained in the span.

Source reference. Original lemma 01R8: sources/stacks-chow/tags.txt, source lines 2293–2293. Navigation pointer for \ref{morphisms-lemma-quasi-compact-scheme-theoretic-image} used inside the counted span (line 2042); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 1993); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $f : X \to Y$ be a proper morphism of schemes which are locally of finite type over S.*

(1) *Let $Z \subset X$ be a closed subscheme with $\dim_\delta(Z) \leq k$. Then*
$$f_*[Z]_k = [f_*\mathcal{O}_Z]_k.$$

(2) *Let $\mathcal{F}$ be a coherent sheaf on X such that $\dim_\delta(Supp(\mathcal{F})) \leq k$. Then*
$$f_*[\mathcal{F}]_k = [f_*\mathcal{F}]_k.$$

Note that the statement makes sense since $f_*\mathcal{F}$ and $f_*\mathcal{O}_Z$ are coherent $\mathcal{O}_Y$-modules by Cohomology of Schemes, Proposition 02O5.

Proof. Part (1) follows from (2) and Lemma 02QY. Let $\mathcal{F}$ be a coherent sheaf on X. Assume that $\dim_\delta(\mathrm{Supp}(\mathcal{F})) \leq k$. By Cohomology of Schemes, Lemma 01Y5 there exists a closed subscheme $i : Z \to X$ and a coherent $\mathcal{O}_Z$-module $\mathcal{G}$ such that

$i_*\mathcal{G} \cong \mathcal{F}$ and such that the support of $\mathcal{F}$ is Z. Let $Z' \subset Y$ be the scheme theoretic image of $f|_Z : Z \to Y$. Consider the commutative diagram of schemes

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ {\scriptstyle f|_Z}\downarrow & & \downarrow{\scriptstyle f} \\ Z' & \xrightarrow{i'} & Y \end{array}$$

We have $f_*\mathcal{F} = f_*i_*\mathcal{G} = i'_*(f|_Z)_*\mathcal{G}$ by going around the diagram in two ways. Suppose we know the result holds for closed immersions and for $f|_Z$. Then we see that

$$f_*[\mathcal{F}]_k = f_*i_*[\mathcal{G}]_k = (i')_*(f|_Z)_*[\mathcal{G}]_k = (i')_*[(f|_Z)_*\mathcal{G}]_k = [(i')_*(f|_Z)_*\mathcal{G}]_k = [f_*\mathcal{F}]_k$$

as desired. The case of a closed immersion is straightforward (omitted). Note that $f|_Z : Z \to Z'$ is a dominant morphism (see Morphisms, Lemma 01R8). Thus we have reduced to the case where $\dim_\delta(X) \leq k$ and $f : X \to Y$ is proper and dominant.

Assume $\dim_\delta(X) \leq k$ and $f : X \to Y$ is proper and dominant. Since f is dominant, for every irreducible component $Z \subset Y$ with generic point η there exists a point $\xi \in X$ such that $f(\xi) = \eta$. Hence $\delta(\eta) \leq \delta(\xi) \leq k$. Thus we see that in the expressions

$$f_*[\mathcal{F}]_k = \sum n_Z[Z], \quad \text{and} \quad [f_*\mathcal{F}]_k = \sum m_Z[Z].$$

whenever $n_Z \neq 0$, or $m_Z \neq 0$ the integral closed subscheme Z is actually an irreducible component of Y of δ-dimension k. Pick such an integral closed subscheme $Z \subset Y$ and denote η its generic point. Note that for any $\xi \in X$ with $f(\xi) = \eta$ we have $\delta(\xi) \geq k$ and hence ξ is a generic point of an irreducible component of X of δ-dimension k as well (see Lemma 02QT). Since f is quasi-compact and X is locally Noetherian, there can be only finitely many of these and hence $f^{-1}(\{\eta\})$ is finite. By Morphisms, Lemma 02NW there exists an open neighbourhood $\eta \in V \subset Y$ such that $f^{-1}(V) \to V$ is finite. Replacing Y by V and X by $f^{-1}(V)$ we reduce to the case where Y is affine, and f is finite.

Write $Y = \mathrm{Spec}(R)$ and $X = \mathrm{Spec}(A)$ (possible as a finite morphism is affine). Then R and A are Noetherian rings and A is finite over R. Moreover $\mathcal{F} = \widetilde{M}$ for some finite A-module M. Note that $f_*\mathcal{F}$ corresponds to M viewed as an R-module. Let $\mathfrak{p} \subset R$ be the minimal prime corresponding to $\eta \in Y$. The coefficient of Z in $[f_*\mathcal{F}]_k$ is clearly $\mathrm{length}_{R_\mathfrak{p}}(M_\mathfrak{p})$. Let $\mathfrak{q}_i$, $i = 1, \ldots, t$ be the primes of A lying over $\mathfrak{p}$. Then $A_\mathfrak{p} = \prod A_{\mathfrak{q}_i}$ since $A_\mathfrak{p}$ is an Artinian ring being finite over the dimension zero local Noetherian ring $R_\mathfrak{p}$. Clearly the coefficient of Z in $f_*[\mathcal{F}]_k$ is

$$\sum\nolimits_{i=1,\ldots,t} [\kappa(\mathfrak{q}_i) : \kappa(\mathfrak{p})]\mathrm{length}_{A_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})$$

Hence the desired equality follows from Algebra, Lemma 02M0. □